

# Yutong Han

Ph.D. Applicant Portfolio



## Visualizing Complexity: Enhancing Trust and Agency in Sociotechnical Systems

*Research Interests: Human-Centered AI, Visual Analytics, Information Accessibility, Algorithmic Transparency*

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Portfolio: <https://yutonghan19951123.github.io>

# Content



2013 - 2017

**China University of Political  
Science and Law**

Bachelor's in Applied Psychology

2018 - 2020

**University of Miami**

Master's in Interactive Media

2021 - 2022

**Caixin Media**

Data Journalist

2022 - current

**Tranalytic Technology**

Front-End & Data Visualization Engineer



1 Intersection AI Agent Interface



2 Pillars: 3D Visualization Framework



3 Auto-Visualization Workspace



4 Which Winter Olympic Sport Fits You



5 Visualizing Wastewater & Ecology



# Intersection AI Agent Interface

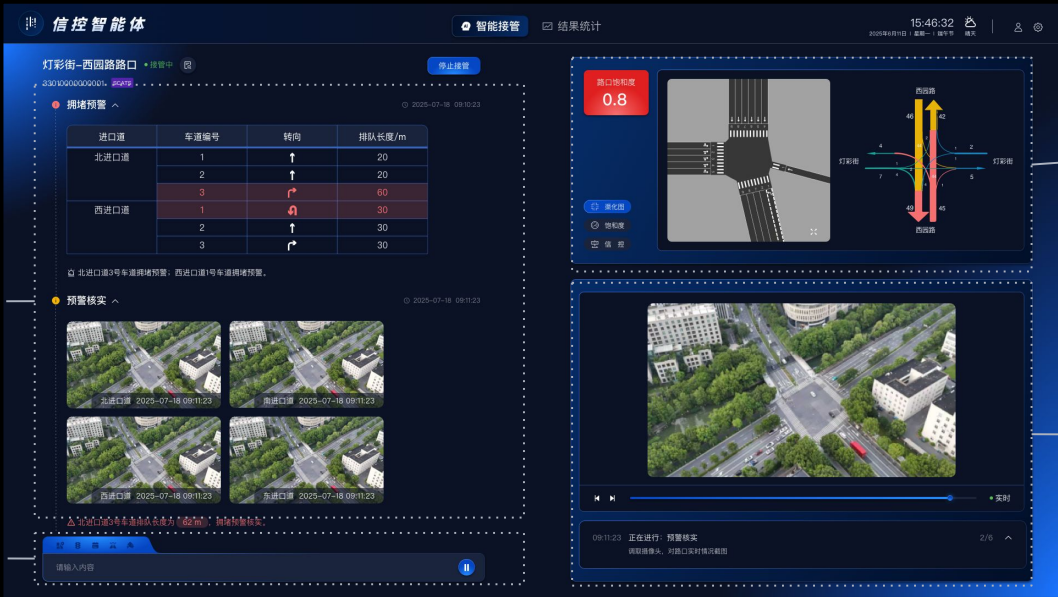
## A Visual-First Human-in-the-Loop System for Adaptive Traffic Signal Control

The interface bridges the trust gap by overlaying semantic ground truth onto CCTV feeds via a custom Canvas engine, using dynamic bounding boxes and sequential saturation zones. Combining this visual auditability with LLM-driven natural language steering transforms opaque signal optimization into a transparent, collaborative process, allowing operators to instantly validate AI logic and execute expert strategies with confidence.

Decision Loop

LLM Interaction

- Duration** July 9 to September 6, 2025
- Role** Lead UI/UX Designer & Front-End Engineer
- Tech Stack** Vue 3, HTML5 Canvas, WebSockets, LLM Integration, YOLOv8
- Key Methods** Visual Analytics, Semantic Segmentation Overlay, Prompt Engineering



# From Black Box to "Glass Box": The 5-Step Loop

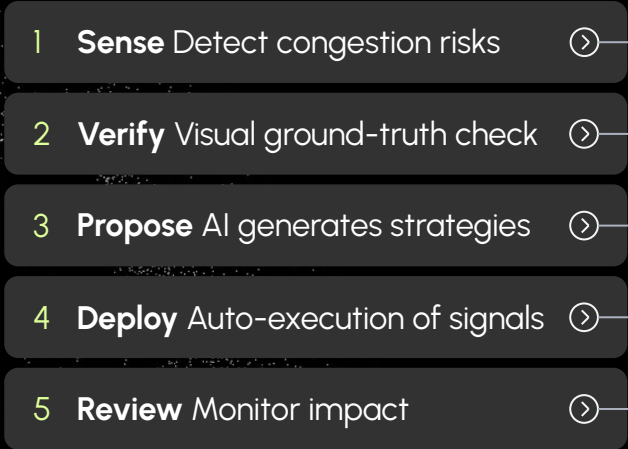
## The Challenge: The Opacity of Automation

Conventional smart traffic systems excel at retrospective analysis of historical data. However, automated intervention in real-time congestion governance remains an opaque "black box." Algorithms adjust signals instantly without explanation, creating two critical issues:

- **Trust Gap:** Operators are hesitant to trust an unauditable AI
- **Cognitive Overload:** Vetting automated decisions requires manually reviewing excessive CCTV and data streams

## The Solution: A Linear Audit Loop

Continuous optimization is now a discrete, verifiable loop: agent working memory is externalized to a linear timeline, enabling autonomous sensing-to-deployment. Operators shift from manual intervention to system supervision via a "glass box."

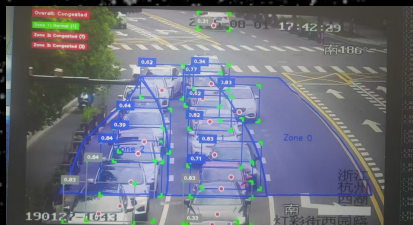


## Timeline Visualization





# Core Innovation



## Canvas Rendering Engine (My Contribution)

Engineered a high-performance HTML5 Canvas overlay system to superimpose semantic data onto the live H.265 video stream, acting as a "visual translator" for backend algorithms:

- **Dynamic Object Tracking:** Implemented real-time, color-coded bounding boxes and tracking IDs to verify YOLO model vehicle counts
- **Zone-Level Saturation Visualization:** Visualized sequential saturation logic by drawing Area of Interest (AOI) polygons, instantly showing if thresholds like "Zone 2 > 2 cars" are met
- **Confidence Exposure:** Rendered model confidence scores (e.g., 0.8) next to targets for real-time assessment of machine vision reliability

# Impact & Reflection

## Operational Impact: Bridging the Expertise Gap

- **Accelerated Onboarding:** The 5-step loop flattens the learning curve for junior operators, speeding up traffic optimization proficiency.
- **Enhanced Auditability:** Visualizing the decision rationale 'black box' via Canvas-based verification allows operators to inspect AI logic against ground truth, increasing trust.
- **Streamlined Supervision:** Transformed the workflow to AI-assisted supervision, enabling operators to manage more intersections simultaneously with reduced cognitive load.

## Research Reflection

Developing this system showed that trust in automation stems from the visibility of its logic, not just algorithmic accuracy.

This project ignited a research passion beyond engineering, prompting me to investigate fundamental trade-offs in visual epistemology:

- To what extent can visual complexity enhance decision-making without causing cognitive overload?
- How does the explicit visualization of algorithmic uncertainty influence human agency and trust in high-stakes environments?

These questions motivate my Ph.D. I aim to move beyond functional tools to define the theoretical principles for trustworthy and effective human-AI collaboration.

# "Pillars": A 3D Interactive Framework for Spatially Coincident Geodata

**Research Goal** Visualizing urban mobility usually leads to a "spatial coincidence" conflict, where inflow and outflow magnitudes occupy the same coordinates.

**Solution** Pillars is a 3D visualization framework developed through a problem-driven design study. Unlike traditional 2D aggregation, it utilizes vertical extrusion and hybrid rendering to disentangle coincident data, achieving real-time performance for 400,000+ daily trip records.

**Duration**

October 15 to November 19, 2025

**Role**

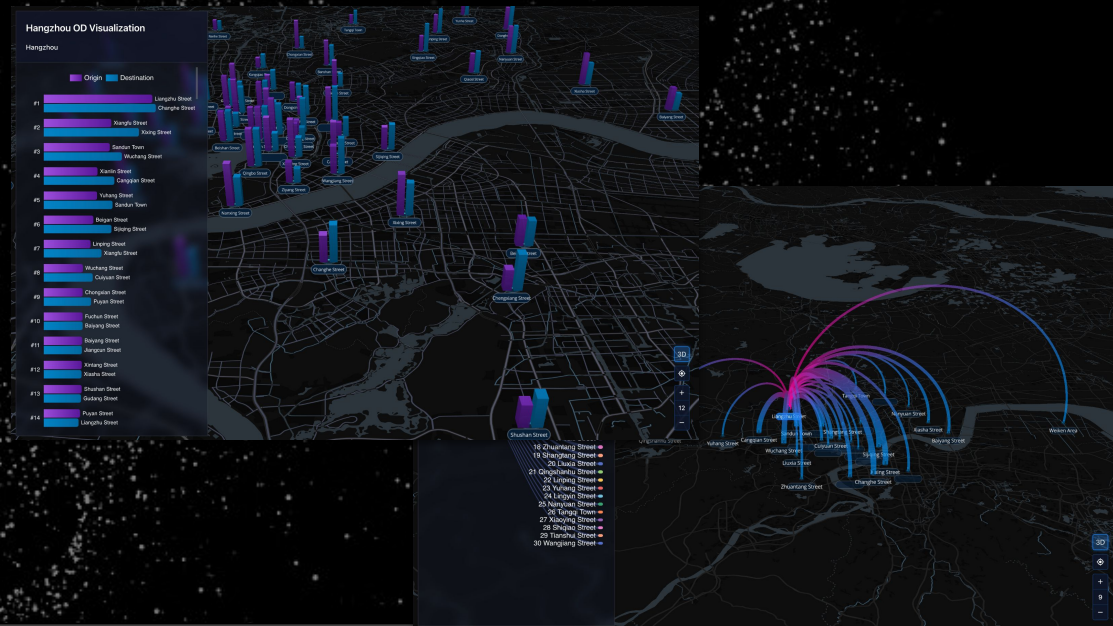
Visualization Lead & Researcher & Front-End Engineer

**Tech Stack**

MapLibre GL JS, deck.gl, HTML5 Canvas, Element-plus

**Key Methods**

Problem-Driven Design Study, Comparative Heuristic Evaluation, Hybrid Rendering Architecture

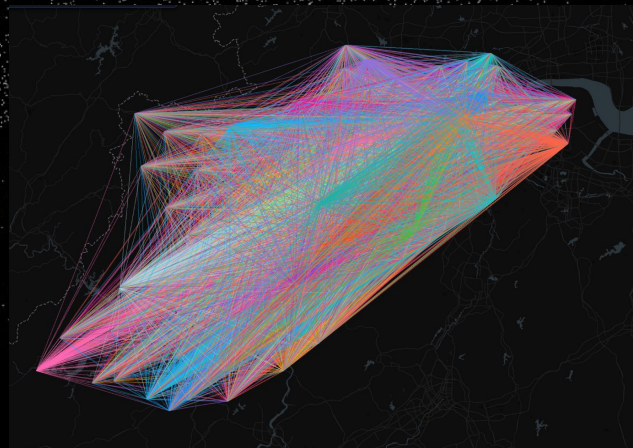




# Problem and Solution

**The Challenge** Topographic Conflict In OD analysis, inflow and outflow magnitudes often map to the exact same (x,y) coordinates.

**Why 2D Fails** As data density increases, standard planar methods collapse into a "spaghetti effect." The massive occlusion masks the underlying geographic truth, making quantitative comparison impossible.



## The Solution: A Linear Audit Loop

### 1 2D Circles

**FAILED**

Severe overlap & spatial instability

### 2 2D Pentagons

**FAILED**

Cognitive load & failed comparison

### 3 Pseudo-3D Markers

**FAILED**

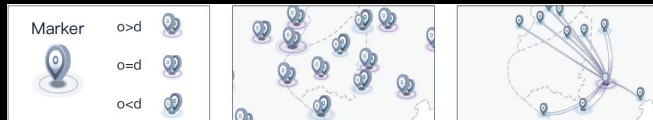
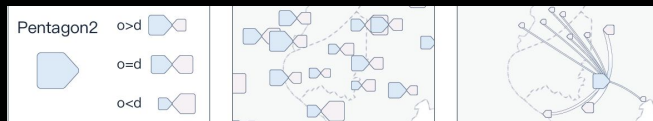
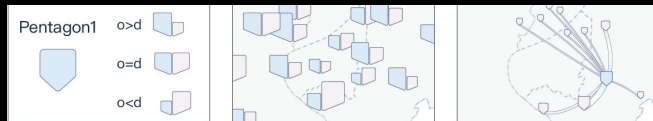
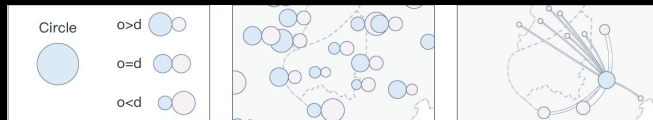
Information loss & perceptual violation

### 4 3D Pillars

Design Version

Overview Map

Detail Map





# Engineering Implementation

## Declarative Middleware

Injected React-style layer management into MapLibre, synchronizing UI and GPU state without manual DOM manipulation.



## Static Topology (MapLibre)

Optimized 386 gradients using a dynamically generated texture atlas, enabling single-draw-call batch rendering for all pillars.

## Dynamic Flow (deck.gl)

Leveraged WebGL Geometry Instancing for thousands of arcs, updating only attributes during interactions to ensure stable 60 FPS transitions.

## Deck.gl Layers

- ArcLayer (Instanced Rendering)
- FillExtrusion (Texture Atlas)
- High frequency updates, acceleration

## MapboxOverlay

- 1 scene - 1 overlay - N layers
- beforeId controls insertion point

## MapLibre GL

- Low-frequency updates
- 60 FPS stable rendering

```
// 10. 创建图层 (用于在场景中绘制所有数据)
export const ArcLayer = refMapLibreLayers({ "ArcLayer": Unknown word,
type: LAYER_TYPE.DECKGL_ARC,
name: LAYER_NAME.DECKGL_ARC, "ArcLayer": Unknown word,
visibility: "hidden",
index: 4,
data: [],
style: {
opacity: 0.2,
},
deckglConfig: {
pinholes: true,
getSourcePositions: (d: any) => d.source,
getTargetPositions: (d: any) => d.target,
getSourceColor: (d: any) => d.source,
getTargetColor: (d: any) => d.target,
getWidth: (d: any) => {
const normalizedValue = Math.min(d.value / 100, 1)
const minWidth = 1
const maxWidth = 10
return minWidth + (maxWidth - minWidth) * normalizedValue
},
widthMapPinholes: 1,
widthMapPinholes: 50,
getHeight: (d: any) => {
const maxHeight = (distance * 0.1) / 111 // 距离为 1 度 = 111 公里
return ((number) => Math.min(Math.PI * (1 + Math.max(maxHeight, 0.01)
),
getZIndex: 0,
getCircle: true,
maxSegments: 100,
},
},
})
```

```
// 11. 创建或更新 deck.gl 图层 (refactored version)
+ Architecture description:
+ 1. Create/update layer instance (including beforeId configuration)
+ 2. Store in layersMap
+ 3. Call updateOverlayLayers to uniformly update MapboxOverlay
+ 4. deck.gl will automatically insert into the correct position based on beforeId
*/
export async function createDeckGLLayer(scene: MapLibreGL.MapType, layer: Layer): Promise<boolean> {
if (!isDeckGLLayerType(layer.type)) {
return false
}

const { name, zIndex } = layer

// 获取当前场景的 overlay
getOrCreateDeckGLLayer(scene)

// 获取当前场景的 layersMap
const layersMap = getDeckGLLayersMap(scene)

// 检查图层是否已存在
const existingLayer = layersMap.get(name)

// 创建 deck.gl 图层实例 (包含 beforeId)
const deckLayer = createDeckGLLayer(scene, layer)

if (!existingLayer) {
return false
}

if (existingLayer) {
// 更新现有图层
const oldZIndex = getLayerZIndex(scene, name) ?? 0

// 更新图层实例
layersMap.set(name, deckLayer)

// 如果 z-index 变化, 更新映射
if (oldZIndex !== zIndex) {
updateZIndexInMap(scene, name, zIndex)
}

// 更新 overlay 的所有图层 (deck.gl 会根据 beforeId 自动调整位置)
updateOverlayLayers(scene)
} else {
// 创建新图层
try {
// 存储图层实例
layersMap.set(name, deckLayer)

// 注册 z-index
registerLayerZIndex(scene, layer)

// 更新 overlay 的所有图层 (deck.gl 会根据 beforeId 自动插入到正确位置)
updateOverlayLayers(scene)
} catch (error) {
return false
}
}

return true
}
```

Live Demo Included: <https://od-visualization-beta.vercel.app/>, Demo Code: <https://github.com/yutonghan19951123/od-visualization>

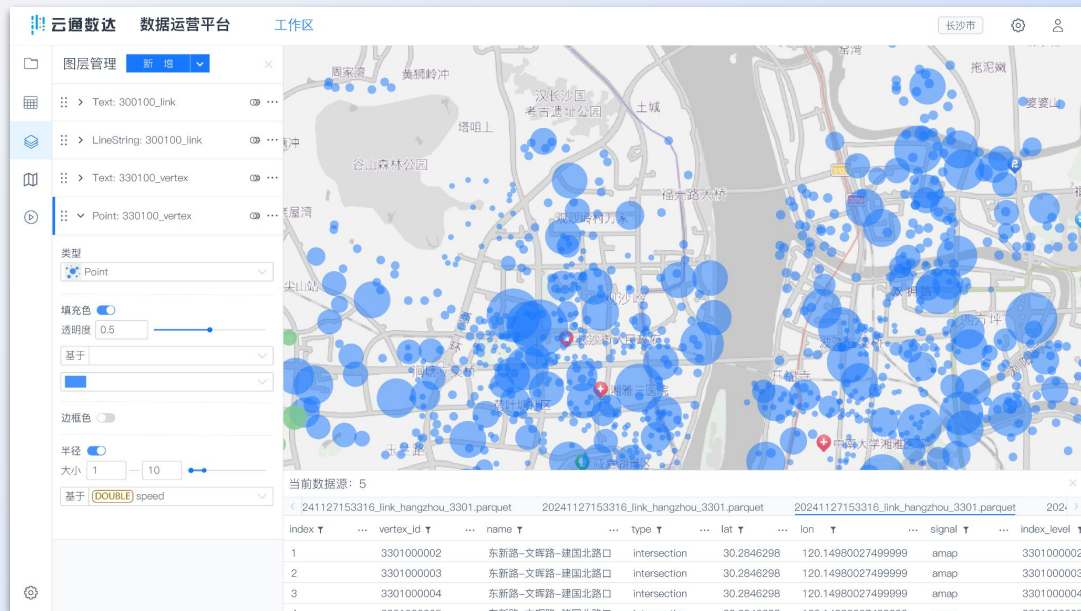
# Auto-Visualization for Geospatial Workflows

## A Decoupled, Declarative Infrastructure for Instant Visual Analytics

I architected a standalone, decoupled workspace to automate the "last mile" of traditional geospatial workflows, which required manual bridging between raw data and visualization tools.

The system uses type-driven visual inference to automatically translate raw dimension tables into optimal visual encodings (e.g., Varchar → Color Category, Numeric → Radius), enabling analysts to explore multiple geographical data without writing code.

|             |                                                              |
|-------------|--------------------------------------------------------------|
| Duration    | December 10, 2024, to February 13, 2025                      |
| Role        | Lead Architect & Front-End Engineer                          |
| Tech Stack  | MapLibre GL JS, Vue 3, Vite, Tailwind CSS, Element-plus      |
| Key Methods | Type-Driven Inference, Progressive Rendering, Layer Stacking |



# Automating the "Last Mile": From Raw Data to Visual Insight

## The Manual Bottleneck

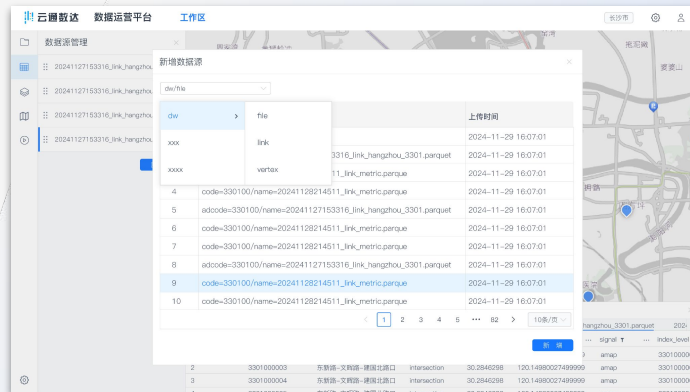
Traditional data ops store vast geospatial data in decoupled formats (CSV, Parquet, API). Visualization demands a rigid, manual pipeline: Query → Export → Manual GIS Configuration, creating high barriers to entry and a slow feedback loop.

## Type-Driven Inference

I developed an inference engine that generates optimal encodings by inspecting data schema on the fly, eliminating manual styling. Dimensions (Varchar) map to categorical palettes, and measures (Double) map to radius/opacity.

The workspace treats visualizations as composable layers. Users can stack distinct datasets (e.g., Road Network + POI + Population) and dynamically switch rendering modes.

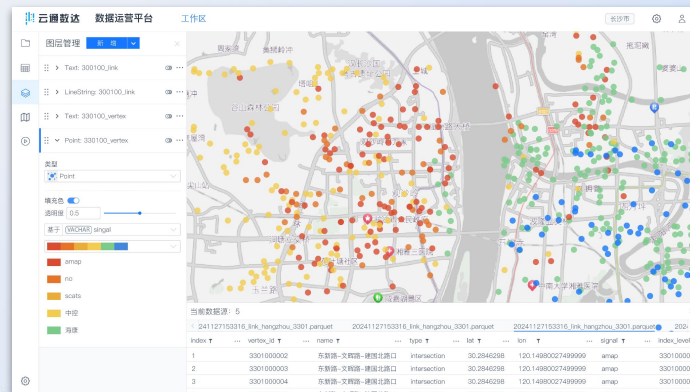
## Choose data source in Workspace



## Multiple options for layer style



## Different bubble color according to varchar field



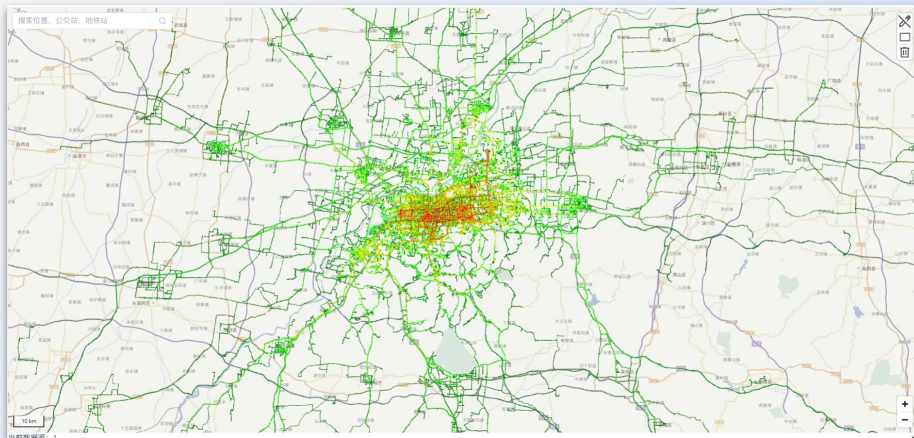
# Engineering Performance

## Optimization for Scale: Precision & Efficiency

To achieve 60 FPS performance with city-scale datasets, I implemented a multi-tier optimization strategy:

- **Spatial Indexing & BBOX Query:** Uses dynamic Bounding Box to limit data requests to the current viewport, cutting data transmission by over 80%.
- **Incremental Loading Strategy:** A smart watcher system loads data incrementally, fetching only the delta when new sources are added, avoiding redundant re-fetching.
- **Debounced Interaction:** Map interactions (pan/zoom) are debounced (1000 ms), reducing invalid API calls by 90% during rapid navigation.
- **Data-View Separation:** Raw query results are decoupled into separate Shape (Geometry) and Text (Label) buffers, allowing independent, on-demand style updates without reprocessing the entire dataset.

*Over 50,000 geohash rect data displaying grid-based heatmap*



**Video Demo Online:** <https://yutonghan19951123.github.io/blog/2025-2-2-data-operation-platform-workspace#outcome>



# Interactive Narrative: Which Winter Olympic Sport Fits You?

## Gamifying Data Journalism for Public Engagement

As the Beijing 2022 Winter Olympics approached, I initiated and led a data journalism project to connect the public with niche winter sports. Originally conceived as a biometric matching tool, the project evolved into a personality-driven interactive game based on user research.

I self-taught character animation (After Effects) to create immersive motion graphics, successfully transforming dry athletic data into an engaging narrative experience.

**Duration**

November 1, 2021, to January 13, 2022

**Role**

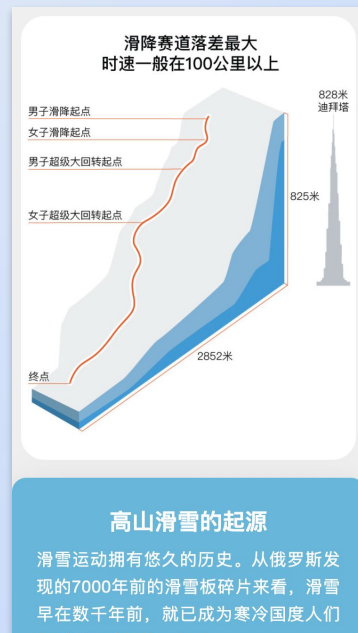
Project Initiator, Lead Designer & Developer

**Tech Stack**

Adobe After Effects (Character Animation), Illustrator, jQuery

**Key Methods**

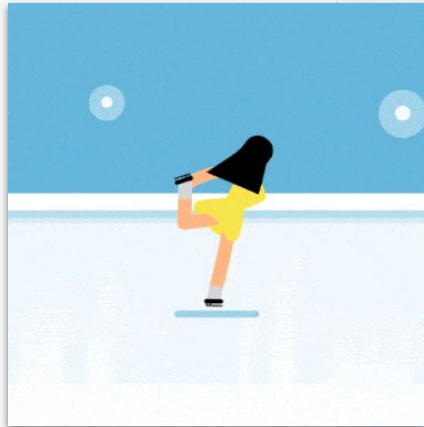
Motion Design, Storytelling, User Research



# Motion Design & Learning

## Autodidactic Motion Design

To make niche sports relatable, I developed character animation skills via an intensive two-month self-study (Skillshare/YouTube) focusing on rigging (Duik) and motion cycles. This allowed me to move from basic displacement to complex movements (like figure skating spins), creating a lively visual language that softened the technicality of winter sports.



# The Product Experience

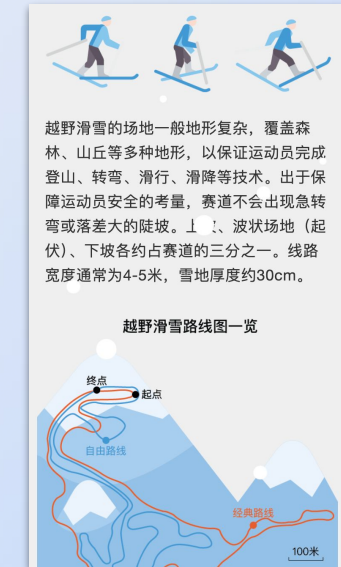
## Users navigate a series of psychological questions

The result page is serving as a data infographic; each page acts as a 'micro-encyclopedia' for the Olympics, detailing the sport's rules, history, and key metrics.

### Gamified Interaction



### Personalized Match



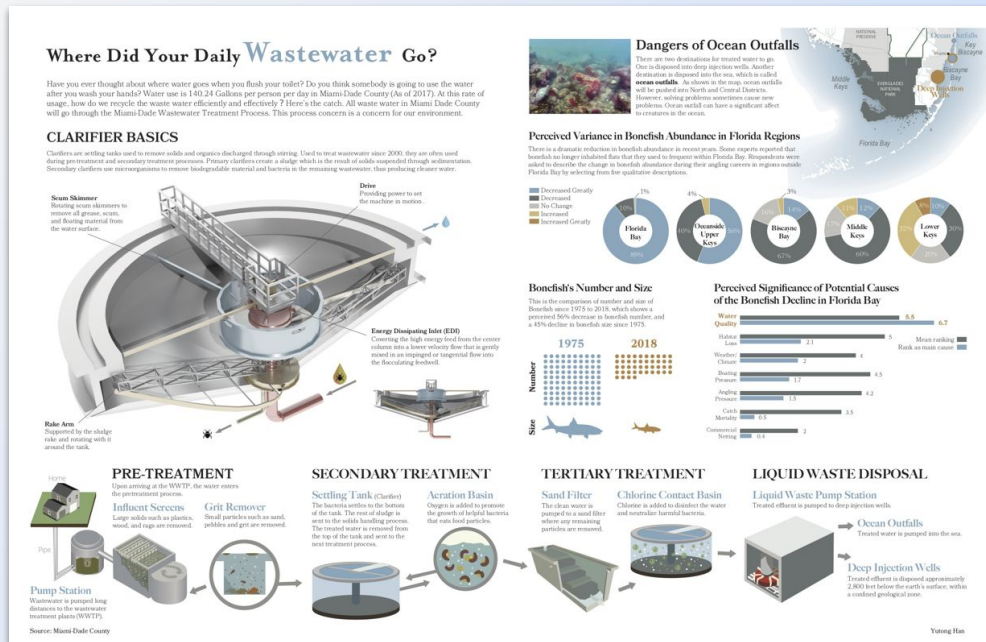
**Website Link Included:** <https://datanews.caixin.com/interactive/2022/winter-olympics-game/>

# Where did your daily wastewater go

## Explaining the Environmental Impact of Ocean Outfalls via 3D Simulation

Investigating Miami-Dade County's wastewater journey and its ecological link to bonefish decline. I synthesized environmental reports and engineering blueprints, using Autodesk Maya to create high-fidelity 3D models of treatment facilities (e.g., clarifiers), translating complex civil engineering into accessible public visual narratives.

**Duration** January 31, 2019, to March 7, 2019  
**Role** Researcher, 3D Modeler, Visual Designer  
**Tech Stack** Autodesk Maya, Adobe Illustrator, Flourish  
**Key Methods** Scientific Visualization, 3D Modeling, Narrative Design



# Thank you

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Portfolio: <https://yutonghan19951123.github.io>